7

Strategic Trade Models

In the sequential trade framework, there are many informed agents, but each can trade only once and only if he or she is "drawn" as the arriving trader. Furthermore, if order size is a choice variable, the informed agent will always trade the largest quantity. The Kyle (1985) model, discussed in this chapter, differs in both respects. In the Kyle model, there is a single informed trader who behaves strategically. She sets her trade size taking into account the adverse price concession associated with larger quantities. Furthermore, She can, in the multiple-period version of the model, return to the market, spreading out her trades over time.

The practice of distributing orders over time to minimize trade impact is perhaps one of the most common strategies used in practice. With decimalization and increased fragmentation of trading activity, market participants have fewer opportunities to easily trade large quantities. In the present environment, therefore, order-splitting strategies are widely used by all sorts of traders (uninformed as well as informed). Although the Kyle model allows for strategic trade, whereas the sequential trade models do not, it is more stylized in some other respects. There is no bid and ask, for example; all trades clear at an informationally efficient price.

7.1 The Single-Period Model

The terminal security value is $v \sim N(p_0, \Sigma_0)$. There is one informed trader who knows v and enters a demand x. Liquidity ("noise") traders submit a net order flow $u \sim N(0, \sigma_u^2)$, independent of v. The market maker (MM) observes the total demand $y = x + u$ and then sets a price, p. All of the trades are cleared at p. If there is an imbalance between buyers and sellers, the MM makes up the difference.

Nobody knows the market clearing price when they submit their orders. Because the liquidity trader order flow is exogenous, there are really only two players we need to concentrate on: the informed trader and the MM. The informed trader wants to trade aggressively, for example, buying a large quantity if her information is positive. But the MM knows that if he sells into a large net customer buy, he his likely to be on the wrong side of the trade. He protects himself by setting a price that is increasing in the net order flow. This acts as a brake on the informed trader's desires: If she wishes to buy a lot, she'll have to pay a high price. The solution to the model is a formal expression of this trade-off.

We first consider the informed trader's problem (given a conjectured MM price function) and then show that the conjectured price function is consistent with the informed trader's optimal strategy. The informed trader conjectures that the MM uses a linear price adjustment rule $p = \lambda y + \mu$ where y is the total order flow: $y = u + x$. λ is an inverse measure of liquidity. The informed trader's profits are $\pi = (v - p)x$. Substituting in for the price conjecture and y yields $\pi = x[v - \lambda(u + x) - \mu]$. The expected profits are $E\pi = x(v - \lambda x - \mu)$. In the sequential trade models, an informed trader always makes money. This is not true here. For example, if the informed trader is buying $(x > 0)$, it is possible that a large surge of uninformed buying $(u \gg 0)$ drives the $\lambda(u + x) + \mu$ above v. The informed trader chooses x to maximize $E\pi$, yielding $x = (v - \mu)/2\lambda$. The second-order condition is $\lambda > 0$.

The MM conjectures that the informed trader's demand is linear in v: $x = \alpha + \beta v$. Knowing the optimization process that the informed trader followed, the MM can solve for α and β: $(v - \mu)/2\lambda = \alpha + \beta v$ for all v. This implies

$$\alpha = -\mu/2\lambda \quad \text{and} \quad \beta = 1/2\lambda. \tag{7.1}$$

The inverse relation between β and λ is particularly important. As liquidity drops (i.e., as λ rises) the informed agent trades less. Now the MM must figure out $E[v|y]$. This computation relies on the following result, which are also used later in other contexts.

Bivariate normal projection. Suppose that X and Y are bivariate normal random variables with means μ_X and μ_Y, variances σ_X^2 and σ_Y^2, and covariance σ_{XY}. The conditional expectation of Y given X is $E[Y|X = x] = \mu_Y + (\sigma_{XY}/\sigma_X^2)(x - \mu_X)$. Because this is linear in X, conditional expectation is equivalent to projection. The variance of the projection error is $\mathrm{Var}[Y|X = x] = \sigma_Y^2 - \sigma_{XY}^2/\sigma_X^2$. Note that this does not depend on x.

Here, given the definition of the order flow variable and the MM's conjecture about the informed traders behavior, $y = u + \alpha + \beta v$, we have $Ey = \alpha + \beta Ev = \alpha + \beta p_0$, $\mathrm{Var}(y) = \sigma_u^2 + \beta^2 \Sigma_0$, and $\mathrm{Cov}(y, v) = \beta \Sigma_0$.

Using these in the projection results gives

$$E[v|y] = p_0 + \frac{\beta(y - \alpha - \beta p_0)\Sigma_0}{\sigma_u^2 + \beta^2\Sigma_0} \quad \text{and}$$

$$\mathrm{Var}[v|y] = \frac{\sigma_u^2\Sigma_0}{\sigma_u^2 + \beta^2\Sigma_0}. \tag{7.2}$$

This must equal $p = \lambda y + \mu$ for all values of y, so

$$\mu = \frac{\alpha\beta\,\Sigma_0 - \sigma_u^2 p_0}{\sigma_u^2 + \beta^2\Sigma_0} \quad \text{and} \quad \lambda = \frac{\beta\,\Sigma_0}{\sigma_u^2 + \beta^2\Sigma_0}. \tag{7.3}$$

Solving Eqs. (7.1) and (7.3) yields:

$$\alpha = p_0\sqrt{\frac{\sigma_u^2}{\Sigma_0}}; \quad \mu = p_0; \quad \lambda = \frac{1}{2}\sqrt{\frac{\Sigma_0}{\sigma_u^2}}; \quad \text{and} \quad \beta = \sqrt{\frac{\sigma_u^2}{\Sigma_0}}. \tag{7.4}$$

7.1.1 Discussion

Both the liquidity parameter λ and the informed trader's order coefficient β depend only on the value uncertainty Σ_0 relative to the intensity of noise trading σ_u^2.

The informed trader's expected profit is:

$$E\pi = \frac{(v - p_0)^2}{2}\sqrt{\frac{\sigma_u^2}{\Sigma_0}}. \tag{7.5}$$

This is increasing in the divergence between the true value and the unconditional mean. It is also increasing in the variance of noise trading. We can think of the noise trading as providing camouflage for the informed trader. This is of practical importance. All else equal, an agent trading on inside information will be able to make more money in a widely held and frequently traded stock (at least, prior to apprehension). How much of the private information is impounded in the price? Using the expression for β in equation (7.2) gives $\mathrm{Var}[v|p] = \mathrm{Var}[v|y] = \Sigma_0/2$. That is, half of the insider's information gets into the price. This does not depend on the intensity of noise trading.

The essential properties of the Kyle model that make it tractable arise from the multivariate normality (which gives linear conditional expectations) and a quadratic objective function (which has a linear first-order condition). The multivariate normality can accommodate a range of modifications. The following exercises explore modifications to the model that still fit comfortably within the framework.

Exercise 7.1 (Partially informed noise traders) The noise traders in the original model are pure noise traders: u is independent of v.

Consider the case where the "uninformed" order flow is positively related to the value: $\mathrm{Cov}(u, v) = \sigma_{uv} > 0$. We might think of u as arising from partially informed traders who do not behave strategically. Proceed as before. Solve the informed trader's problem; solve the MM's problem; solve for the model parameters $\{\alpha, \beta, \mu, \lambda\}$ in terms of the inputs $\{\sigma_u^2, \Sigma_0, \sigma_{uv}\}$. Interpret your results. Show that when $\mathrm{Corr}(u, v) = 1$, the price becomes perfectly informative.

Exercise 7.2 (Informed trader gets a signal instead of complete revelation) The informed trader in the basic model has perfect information about v. Consider the case where she only gets a signal s. That is, she observes $s = v + \varepsilon$ where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$, independent of v. Solve for the model parameters $\{\alpha, \beta, \mu, \lambda\}$ in terms of the inputs, σ_u^2, Σ_0, and σ_ε^2. Interpret your results. Verify that when $\sigma_\varepsilon^2 = 0$, you get the original model solutions.

Exercise 7.3 Around 1985, a Merrill Lynch broker noticed a pattern of profitable trades originating from a Bahamas branch of Bank Leu and began to piggyback his own orders on the Bank Leu flow. Suppose that when the informed trader in the basic model puts in an order x, her broker simultaneously puts in an order x, with γx, with $\gamma > 0$. Solve for the model parameters $(\alpha, \beta, \mu, \lambda)$ in terms of the inputs, σ_u^2, Σ_0, and γ. It turned out that the Bank Leu orders originated from a New York investment banker, Dennis Levine, who subsequently pleaded guilty to insider trading (see Stewart (1992)).

7.2 Multiple Rounds of Trading

A practical issue in market design is the determination of when trading should occur. Some firms on the Paris Bourse, for example, trade in twice-per-day call auctions, others continuously within a trading session. What happens in the Kyle model as we increase the number of auctions, ultimately converging to continuous trading?

We will consider the case of N auctions that are equally spaced over a unit interval of time. The time between auctions is $\Delta t = 1/N$. The auctions are indexed by $n = 1, \ldots N$. The noise order flow arriving at the nth auction is defined as Δu_n. This is distributed normally, $\Delta u_n \sim N(0, \sigma_u^2 \Delta t)$ where σ_u^2 has the units of variance per unit time. The use of the difference notation facilitates the passage to continuous time. The equilibrium has the following properties.

- The informed trader's demand is $\Delta x_n = \beta_n (v - p_{n-1}) \Delta t_n$, where β_n is trading intensity per unit time.
- The price change at the nth auction is $p_n = p_{n-1} + \lambda_n (\Delta x_n + \Delta u_n)$.
- Market efficiency requires $p_n = E[v|y_n]$ where y_n is the cumulative order flow over time.

In equilibrium the informed trader "slices and dices" her order flow, distributing it over the N auctions. This corresponds to the real-world practice of splitting orders, which is used by traders of all stripes, not just informed ones.

Recall that in the sequential trade models, orders are positively autocorrelated (buys tend to follow buys). Does a similar result hold here? Is autocorrelation a consequence of the informed trader's strategic behavior? There is an argument supporting this conjecture that is intuitive and compelling at first glance. The informed trader splits her orders over time, and so tends to trade on the same side of the market. If her information is favorable $v > p_0$, she will be buying on average; with negative information, she'll be selling. This establishes a pattern of continuation in her orders, and as a result $\mathrm{Corr}(\Delta x_i, \Delta x_j) > 0$ for any pair of auctions i and j we might care to examine. Total order flow, being the sum of her demand and that of the uninformed, is thus the sum of one series that is positively autocorrelated and one that is uncorrelated. We'd expect the positive autocorrelation to persist in the total, diluted perhaps by the uncorrelated noise, but present nonetheless.

In fact, however, total order flow is uncorrelated. Correlation would imply that order flow is at least partially predictable. Since order flow and price changes are linearly related (the λ_n are known at the start of the day), predictability in the former is equivalent to predictability in the latter. Autocorrelation in price changes is ruled out by market efficiency. More formally, the conditional expectation of the incremental order flow is:

$$E[\Delta y_n | y_{n-1}] = E[\Delta u_n + \Delta x_n | y_{n-1}] = E[\beta_n (v - p_{n-1}) \Delta t | y_{n-1}] = 0.$$

The first equality is definitional; the second holds because the noise order flow has zero expectation. The third equality reflects the market efficiency condition that $E[v|y_{n-1}] = p_{n-1}$. From a strategic viewpoint, the informed trader hides behind the uninformed order flow. This means that she trades so that the MM can't predict (on the basis of the net order flow) what she will do next.

Figure 7.1 depicts the market depth parameter over time as a function of the number of auctions. In the sequential trade models, manipulation by an uninformed trader is not generally profitable (section 5.6). A similar result holds here. Huberman and Stanzl (2004) show that the linear price schedules of the Kyle model do not allow manipulation and also that *only* linear price schedules have this property.

7.3 Extensions and Related Work

The Kyle model lies near the center of a large family of theoretical analyses. The following list is suggestive of the directions, but far from comprehensive. Back (1992) develops continuous time properties. Back and

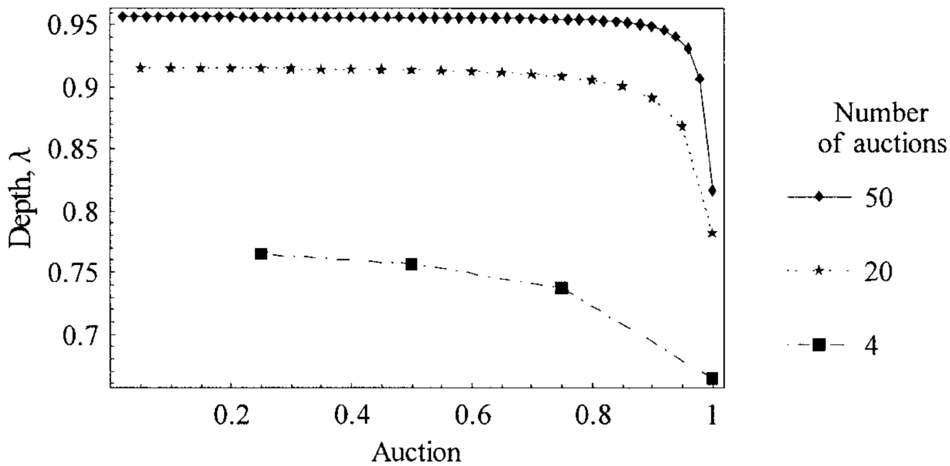

Figure 7.1. Market impact and number of auctions.

Baruch (2004) explore the links between the strategic and sequential trade models. Notably, in their common framework, the informed agent trades randomly in continuous time. Admati and Pfleiderer (1988) incorporate strategic uninformed traders (also see Foster and Viswanathan 1990). Subrahmanyam (1991a) considers risk-averse market makers and informed traders. Spiegel and Subrahmanyam (1992) model the uninformed agents as rational risk-averse hedgers. Analyses that feature multiple securities include Caballe and Krishnan (1994) and Subrahmanyam (1991b) (also see Admati (1985)). Subrahmanyam considers the case of a multiple security market that includes a basket security, for example, a stock index futures contract or an index exchange traded fund. The diversification in an index greatly mitigates firm-specific information asymmetries, making index securities generally cheaper to trade. Holden and Subrahmanyam (1992, 1994), Foster and Viswanathan (1996), and Back, Cao, and Willard (2000) consider multiperiod models with multiple informed traders.